NON-HOMOGENEOUS ODE

1. Solve $y'' - 3y' + 2y = e^{2x}$.
2. Solve $y'' + 4y = 8 \cos 2x$.
3. Solve $y'' - y = x e^{x}$.

POWER SERIES ODE

1. Solve $y'' + y = 0$ using a power series.
2. Solve $y'' - x y = 0$.
3. Solve $y'' + x^2 y = 0$.

NON-HOMOGENEOUS ODE — SOLUTIONS (STEP-BY-STEP)

1. $y'' - 3y' + 2y = e^{2x}$

Homogeneous:

$r^2 - 3r + 2 = 0 \rightarrow r=1,2 \rightarrow y_c = C1\, e^{x} + C2\, e^{2x}$

Particular: e^{2x} is a root → try $y_p = A\, x\, e^{2x}$

Solve → $A = 1$

Final: $y = C1\, e^{x} + C2\, e^{2x} + x\, e^{2x}$

2. $y'' + 4y = 8 \cos 2x$

Homogeneous: $r^2 + 4=0 \rightarrow y_c = C1 \cos 2x + C2 \sin 2x$

Resonance → try $y_p = x(A \sin 2x)$

Solve → $A = 2$

Final: $y = C1 \cos 2x + C2 \sin 2x + 2x \sin 2x$

3. $y'' - y = x\, e^{x}$

Homogeneous: $y_c = C1\, e^{x} + C2\, e^{-x}$

Try: $y_p = e^{x}(Ax^2 + Bx)$

Solve: $A = 1/2$, $B = -1$

Final: $y = C1\, e^{x} + C2\, e^{-x} + e^{x}(1/2\, x^2 - x)$

POWER SERIES ODE — SOLUTIONS (STEP-BY-STEP)

1. $y'' + y = 0$

Assume $y = \Sigma\, a_n x^n \rightarrow y'' = \Sigma\, n(n-1)a_n x^{n-2}$

Plug into ODE: $\Sigma\, n(n-1)a_n x^{n-2} + \Sigma\, a_n x^n = 0$

Shift index to match powers: $\Sigma\, (n+2)(n+1)a_{n+2} x^n + \Sigma\, a_n x^n = 0 \rightarrow (n+2)(n+1)a_{n+2} + a_n = 0 \rightarrow a_{n+2} = -a_n / ((n+2)(n+1))$

Even and odd terms → $\cos x$ and $\sin x$.

2. $y'' - x y = 0$

Same series setup: $y = \Sigma\, a_n x^n$ $y'' = \Sigma\, n(n-1)a_n x^{n-2}$

Plugging in: $\Sigma\, n(n-1)a_n x^{n-2} - \Sigma\, a_n x^{n+1} = 0$

Shift indices: $a_{n+3} = a_n / ((n+3)(n+2))$ 3. $y'' + x^2 y = 0$

Series substitution → $a_{n+2} = -a_{n-2}/((n+2)(n+1))$

Two independent series solutions.